



GASTROENTERITIS (FOOD POISONING)

Primary Care Course

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Learning Objectives

By the end of this session you will be able to:

- Define gastroenteritis and food poisoning
- Identify main causes of gastroenteritis
- Explain some mechanisms of gastroenteritis
- Recognize clinical features of gastroenteritis
- Tell some treatment measures for gastroenteritis
- Define preventive measures for gastroenteritis

Definitions

Gastroenteritis is inflammation of the lining of the stomach, small and large intestines. This inflammation is often caused by infection agents and non-infectious causes such as drugs, medications, and chemical toxins.

Food poisoning, also called **foodborne infection**, is an infection or inflammation of gastrointestinal tract that spreads through food or drinks.

Definitions

Gastroenteritis often manifests in symptoms such as diarrhoea, vomiting or both, abdominal pain, and cramps.

Gastroenteritis typically lasts about three days. Adults usually recover without problem, but children, the elderly, and anyone with an underlying disease are more vulnerable to complications such as dehydration.

Incidence of Gastroenteritis

- Gastroenteritis (**GE**) is a common health problem in developing countries. Worldwide, an estimated 1.6 million people die each year of infectious gastroenteritis.
- Of note, developing countries have a higher rate of hospital admissions from GE as compared to developed countries. In particular, it is estimated that every child encounters at least one episode of rotavirus gastroenteritis by the age of 5 years.

Incidence of Gastroenteritis

- Improvements in water sanitation in many parts of the world and the appropriate use of oral rehydration therapy for infants with diarrhea are likely responsible for this decrease.
- Acute GE is a major killer of the very young worldwide; current estimates suggest that acute gastroenteritis affects 3 to 5 billion children below 5 years of age each year, and accounts for 1.5 to 2.5 million deaths per year.

Causes of Gastroenteritis

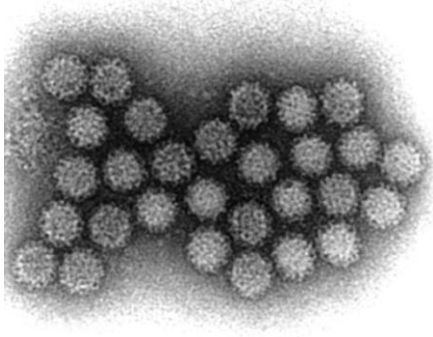
Gastroenteritis is caused by either infection agents or non infection agents.

Infection causes of GE include several agents:

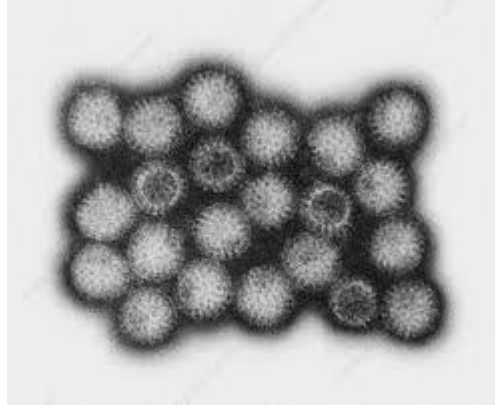
1. Viral infections; most common cause, implicated in nearly 70% of cases.

- Norovirus: Most common cause of gastroenteritis worldwide, especially in adults.
- Rotavirus: Primarily affects infants and young children.
- Astrovirus, and less commonly enteric adenovirus.

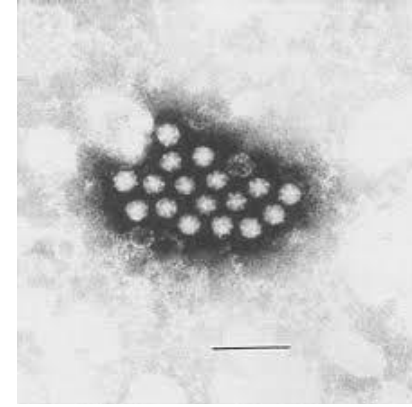
Causes of Gastroenteritis



Norovirus



Rotavirus



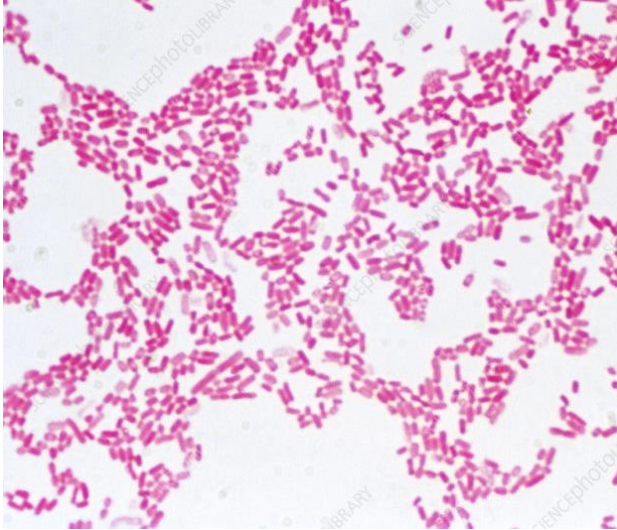
Astrovirus

Causes of Gastroenteritis

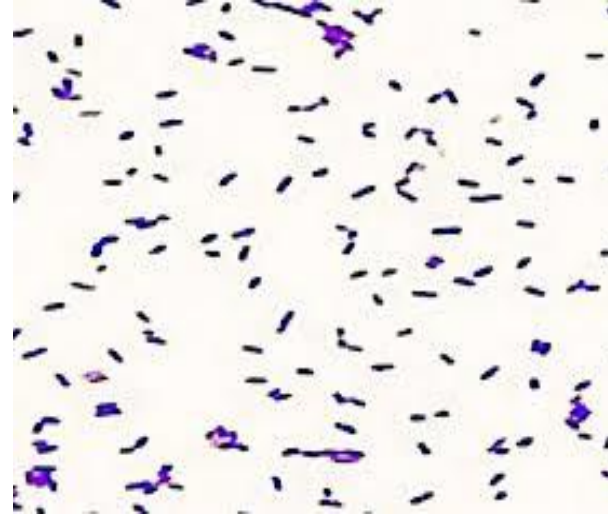
2. Bacterial causes; implicated in 10-20% cases

- *Escherichia coli*: Particularly the enterohemorrhagic strains like *E. coli* O157:H7.
- *Salmonella*, (non typhoid *Salmonella*): Often linked to foodborne outbreaks.
- *Shigella*
- *Campylobacter*: Frequently associated with dysentery-like symptoms.
- *Listeria monocytogenes*

Causes of Gastroenteritis



Escherichia coli



Listeria monocytogenes

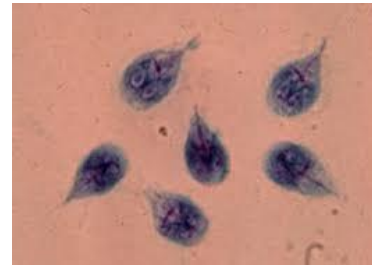
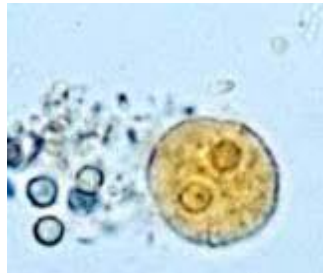
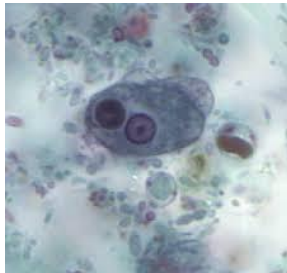
Causes of Gastroenteritis

3. Parasitic Infections: Protozoa are implicated for less than 10% of cases:

- *Giardia lamblia*: Commonly spread through contaminated water.
- *Cryptosporidium* and *Entamoeba histolytica*

Non-Infectious causes:

- Medications: NSAIDs, certain antibiotics.
- Food allergies: Lactose intolerance, gluten sensitivity (celiac disease).



Causes of Gastroenteritis

Important Organisms That Typically Cause Either Watery or Bloody Diarrhea

Organisms That Cause Watery Diarrhea	Organisms That Cause Bloody Diarrhea
Enterotoxigenic <i>Escherichia coli</i> (ETEC)	Shiga toxin-producing <i>Escherichia coli</i> (STEC)
<i>Vibrio cholerae</i>	<i>Shigella</i> species
<i>Staphylococcus aureus</i>	<i>Salmonella enterica</i>
<i>Bacillus cereus</i>	<i>Campylobacter jejuni</i>
<i>Listeria monocytogenes</i>	<i>Clostridium difficile</i>
Norovirus	<i>Yersinia enterocolitica</i>
Rotavirus	<i>Entamoeba histolytica</i>
<i>Giardia lamblia</i>	
<i>Cryptosporidium hominis</i>	

Causes of Gastroenteritis

Benzamin - Mujib Diarrhea stool card



Fig a : Rotavirus Diarrhea



Fig b : Cholera



Fig c : ETEC Diarrhea



Fig d : Dysentery

[Use of 'Diarrhea Stool Card' in acute diarrhea management in under 5 Years children in resource constraints country - ScienceDirect](#)

Pathophysiology of Gastroenteritis

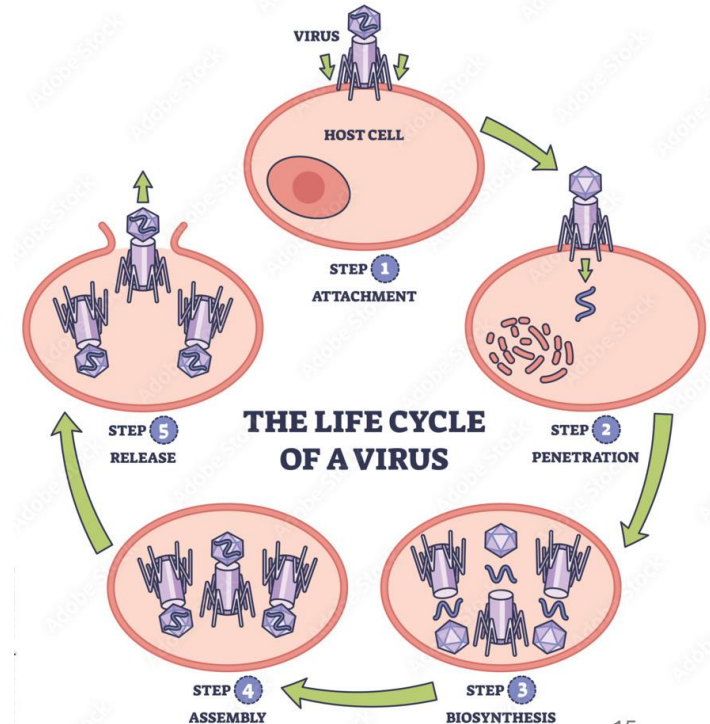
Pathogens or their associated toxins disrupt the normal absorption and secretory processes in the small intestines.

Acute diarrhea is usually caused by preformed exotoxins in food or by the infectious agents in the intestinal. Other may invade intestinal mucosa while other cause pathological effects on mucosal cells.

Pathophysiology of Gastroenteritis

The pathogenesis of gastroenteritis depends on the causative agent:

- Viral pathogens often invade enterocytes in the small intestine, leading to cell death and reduced absorption.



Pathophysiology of Gastroenteritis

- Bacteria may produce preformed toxins (*S. aureus*, *Clostridium perfringens*, *Bacillus cereus*) that disrupt normal fluid absorption.
- Other pathogens that cause non-inflammatory acute diarrhea by enterotoxin production include enterotoxigenic *Escherichia coli* (ETEC) and *Vibrio cholerae*.

Pathophysiology of Gastroenteritis

- Bacteria that cause acute inflammatory diarrhea include *Salmonella*, *Shigella*, *Campylobacter* (via mucosal invasion), Shiga toxin-producing *E. coli* (STEC) such as *E. coli* O157:H7, and *C. difficile* (via cytotoxin production).
- Parasitic agents invade and damage the mucosal lining, impairing nutrient absorption.

Clinical Manifestations of Gastroenteritis

Patients suffering from acute GE typically present with:

- Diarrhea: Ranges from mild to severe, can be watery or bloody.
- Vomiting: Often accompanies diarrhea, especially in viral infections.
- Abdominal Pain: Cramping or general discomfort.
- Systemic Symptoms: Fever, malaise, **dehydration**.
- Blood or mucus in the stool.



Clinical Manifestations of Gastroenteritis

The WHO defines diarrhea as the presence of 3 or more loose or watery stools and an increase in the number of bowel movements in a 24-h period. Diarrhea is considered acute if duration is under 14 days.



Clinical Manifestations of Gastroenteritis

Typically, the symptoms last only two to three days, but some viruses may last up to a week.

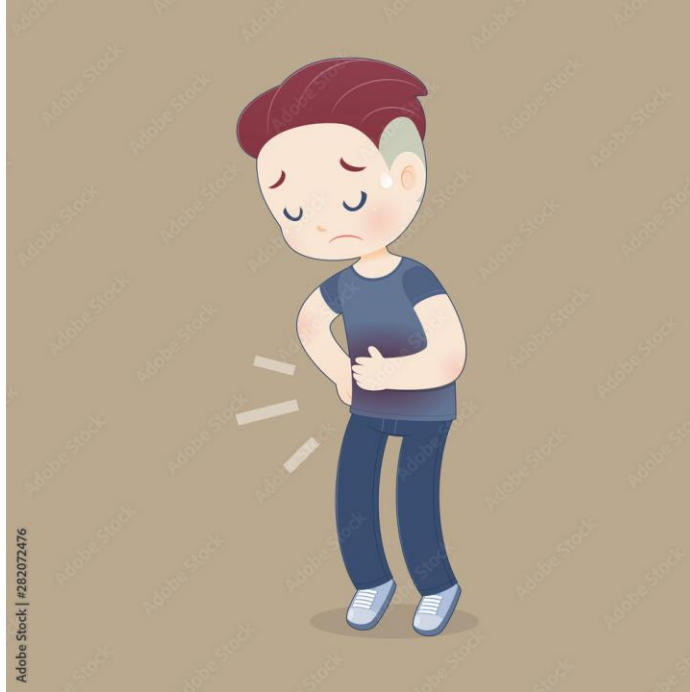
The greatest danger presented by gastroenteritis is **dehydration**. The loss of fluids through diarrhea and vomiting can disturb the body's electrolyte balance, leading to potentially life-threatening problems.

Clinical Manifestations of Gastroenteritis

Characteristics of Watery Diarrhea Compared to Bloody Diarrhea

Characteristics of Watery Diarrhea	Characteristics of Bloody Diarrhea
No red blood cells or white blood cells in stool, i.e., no inflammation	Typically both red blood cells and white blood cells in stool, i.e., inflammatory response
Typically afebrile	Often febrile
Usually large-volume diarrhea	Usually small-volume diarrhea
Infection typically in small intestine	Infection typically in colon

Clinical Manifestations of Gastroenteritis

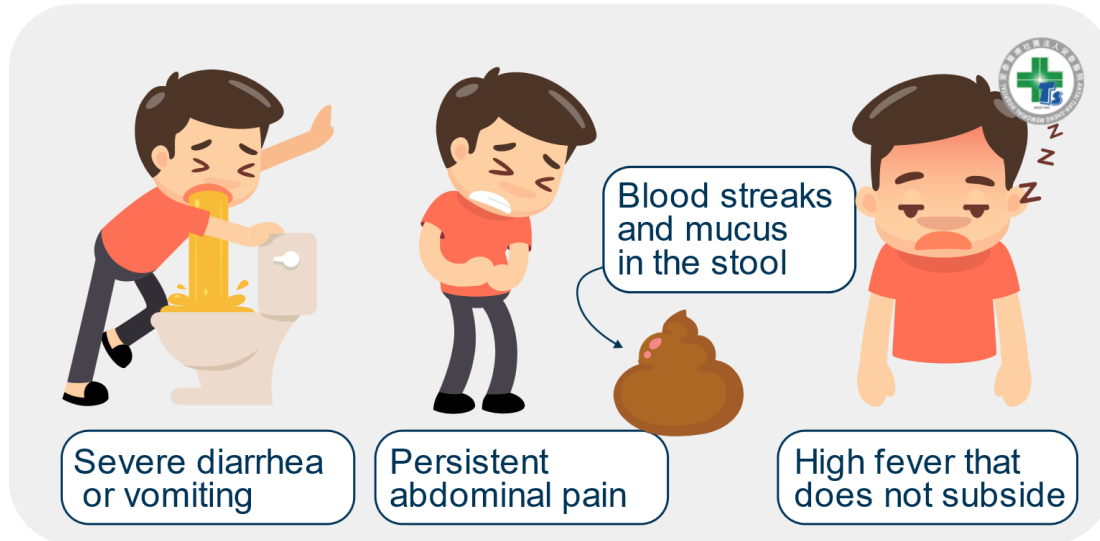


Diagnosis of Gastroenteritis

- Clinical evaluation based on history and physical examination.
- Features of fever, blood or mucus in the diarrhea and severe abdominal pain or swelling cause more concern for diagnosis.
- Unless there is an outbreak affecting several people or complications are encountered in a particular case, identifying the specific cause of the illness is not a priority.

Diagnosis of Gastroenteritis

- However, if identification of the infectious agent is required, a stool sample will be collected and analyzed for the presence of viruses, disease-causing bacteria, or parasites.



Diagnosis of Gastroenteritis

Laboratory Tests:

- Microscopy for stool to detect parasite (*Entamoeba* and *Giardia*) cysts or trophozoites. Stool antigen tests are useful. For *Cryptosporidium hominis* finding cysts in acid-fast stain of stool.
- Food and stool can be tested for toxin as in *S. aureus*, *Bacillus cereus* and *Clostridium difficile*.
- For viral causes, rapid antigen test and stool PCR test are available.

Diagnosis of Gastroenteritis

Laboratory Tests:

- Stool Cultures: Identifies bacterial causes. Using different culture media (selective) to isolate common causative agent. This is useful in *E. coli*, *Shigella*, *Salmonella*, *Campylobacter*, *Yersinia*, and *Listeria*.
- Molecular tests for toxin gene detection using DNA probs.

Diagnosis of Gastroenteritis

Laboratory Tests:

- Blood Tests: Assess severity of dehydration and electrolyte imbalances.
- Imaging: Generally reserved for severe or complicated cases.



Treatment of Gastroenteritis

Several measure on:

1. Rehydration: Most crucial aspect, especially for children and the elderly.

- Oral Rehydration Solutions (ORS): For mild to moderate dehydration.
- Intravenous Fluids: For severe dehydration. Several choices depending on clinical choices.

Treatment of Gastroenteritis

ORS is an oral powder-containing mixture of glucose, sodium chloride, potassium chloride and sodium citrate.

Composition

Sodium Chloride	3.5 g.
Potassium Chloride	1.5 g.
Trisodium Citrate dihydrate	2.9 g.
Glucose Anhydrous	20.0 g.

All for 1 litre of water





Oral Rehydration Solution (ORS)

Flood-contaminated water increases the risk of diarrhoea and vomiting.
ORS prevents severe dehydration caused by frequent diarrhoeal episodes.



Boil one litre water in a clean utensil and allow it to cool down.



Add ORS powder to the water and mix well.



Store the ORS mixture in a clean utensil or bottle.



To make ORS at home,
add six teaspoons sugar
and ½ teaspoon salt in
one litre clean and
boiled water.

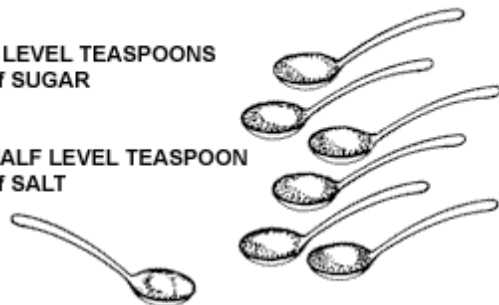
After each episode of vomiting or diarrhoea, give:

- ¼ to ½ cup of ORS solution to children below 2-years
- ½ to one cup to children above 2-years.

Once prepared, consume within 24 hours.

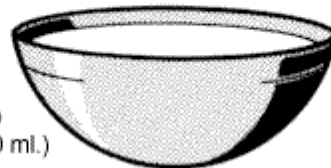
6 LEVEL TEASPOONS
of SUGAR

HALF LEVEL TEASPOON
of SALT



1 LITRE
OF WATER

5 cupfuls
(each cup
about 200 ml.)



BE PREPARED. BE SAFE.



Treatment of Gastroenteritis

Several measure on:

2. Dietary Management:

- Bananas, Rice, Apple sauce, Toast – (easily digestible foods.)
- Avoid fatty foods, and caffeine.



Treatment of Gastroenteritis

3. Pharmacologic Treatment:

- Antiemetics: For severe vomiting.
- Antidiarrheal agents: In selected cases; not recommended for infectious gastroenteritis. For example, Loperamide hydrochloride
- Antibiotics: Only for bacterial infections confirmed by culture. Ciprofloxacin (bacteria), Metronidazole (parasites.)

Treatment of Gastroenteritis

4. Probiotics:

- Bacteria that are beneficial to a person's health, are recommended during the recovery phase of gastroenteritis.
- Specifically, live cultures of *Lactobacillus acidophilus* are said to be effective in soothing the digestive tract and returning the intestinal flora to normal. *L. acidophilus* is found in live culture yogurt.

Prevention of Gastroenteritis

Making sure that food is well-cooked and unspoiled can prevent bacterial gastroenteritis, but this may not be effective against viral gastroenteritis; follow these:

1. Hygiene Measures:

- Handwashing: Critical to prevent spread. Washing with soap before handling food and after using the toilet.
- Clean the toilet and bathroom regularly, especially the toilet seat, door handles and taps.
- Be cautious from close contact with people infected with gastroenteritis.



Prevention of Gastroenteritis

2. Water Safety:

- Avoiding contaminated water sources.
- Drink safe water and only bottled water when travelling especially in developing countries.

3. Food Safety:

- Proper handling and cooking of food.
- Using a separate cutting board for raw meat.
- Avoiding eating of raw meat and fish, unwashed vegetables and exposed food.



Prevention of Gastroenteritis

4. Vaccination:

- Rotavirus vaccine reduces incidence in infants.
- CDC recommends that infants get rotavirus vaccine to protect against rotavirus disease.
- The vaccine contain live attenuated rotavirus .

Prevention of Gastroenteritis

Two rotavirus vaccines are currently licensed for use in infants in the United States:

- RotaTeq® (RV5) has been approved for use since 2006, and is given in three doses at 2 months, 4 months, and 6 months of age.
- Rotarix® (RV1) has been approved for use since 2008, and is given in two doses at 2 months and 4 months of age.
- Both rotavirus vaccines are administered orally, by putting drops in the infant's mouth.



Prevention of Gastroenteritis



COLLEGE
OF
MEDICINE





Non-infectious Gastroenteritis

Non-infectious possible causes of gastroenteritis include:

- Medications, such as:
 - nonsteroidal anti-inflammatory drugs (NSAIDs)
 - antibiotics
 - chemotherapy drugs
 - laxatives
 - digoxin
 - colchicine
- Food intolerances, such as those to lactose and gluten.
- Exposure to toxins such as heavy metals or toxins in contaminated seafood.

Non-infectious Gastroenteritis

- For diagnosis, good history, drug history, direct question may help in diagnosis of non-infectious causes of gastroenteritis.
- Chronicity of the condition and no-response to treatment may suggest a non-infection cause.
- Individual treatment may be followed for these non-infection causes.

Conclusion

- Gastroenteritis remains a significant global health issue, particularly in developing countries where access to clean water and sanitation is limited.
- Early diagnosis and appropriate management by rehydration and restoration of electrolytes are essential to prevent complications.
- Clean drinking water, safe food, good hygiene are the preventive measures for most infection.

References

Review of Medical Microbiology and Immunology 15thEd 2018

Merk Manual. Overview of Gastroenteritis By Jonathan Gottfried